An Exact Analytic Derivation of the Polygon Test Boundary for Online Min-Entropy Testing of Binary Data

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Abstract

This paper replaces the graphical extraction of polygon contours in [1] with a mathematical derivation of the boundary enabling more accurate min-entropy estimation of binary time series random data.

A region of count pairs with asymptotic min-entropy rate at least H is shown to be bounded by exactly two straight lines arising from the all-ones and all-zeros repeating patterns, one straight feasibility line, and one curved edge: an arc of an ellipse arising from the alternating pattern. Closed forms are given for all corner points, for the boundary equations at any entropy threshold, and for a conservative linear chord that replaces the elliptic arc with one-sided error. The geometry changes from a five-sided polygon to a 3 sided triangle (albeit with one edge being the elliptical arc) when $H^* \geq \log_2 \varphi \approx 0.694$, where φ is the golden ratio. Integer comparator constants with conservative rounding are derived for hardware implementation, along with an integer multiplier-based structure that tests the elliptic arc exactly, eliminating the one-sided error rising from a linear approximation of the arc.

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1 Introduction

The paper[1] describes an online entropy test (named the Polygon Test). By defining mathematical relationships between the counts of 1s and 1,1 bit pairs in binary time series data with the min-entropy of the data, a set of polygons is formed on the pattern count plane of 1s and 11s, where each polygon represents iso-entropy contours. Linear inequality tests showing the counts lay within a particular iso-entropy contour allow an efficient hardware or software online entropy test to be implemented that shows in real time if the min-entropy of a binary data sequence from a binary random noise source is above the chosen bound. The polygon is extracted graphically from the iso-entropy plot which is less than perfectly accurate.

This paper replaces the graphical extraction with a mathematical derivation of the boundary.

A two-state Markov model with transition probabilities P_{01} (probability a 0 is followed by a 1) and P_{10} (probability a 1 is followed by a 0) can generate binary data with any chosen mean μ and serial correlation coefficient (SCC), and its min-entropy is computable from the model parameters [1]. That paper further showed that the two counters

$$\textit{count1} = (\text{number of 1s in the first } n-1 \text{ bits}), \qquad \textit{count11} = (\text{number of 11 bit pairs})$$

over an n-bit block carry the same information as the Markov transition probabilities of a two state Markov generator, and that the region of count pairs corresponding to min-entropy above a threshold appears, when plotted, to be a polygon testable with a small number of linear inequality tests — the *polygon test*, deployed as the vendor-defined continuous health test in fifth generation Intel CPU RNGs.

The polygon of [1] was extracted from a rendered contour plot. One edge appeared curved and convex, but it was not established whether it was piecewise linear or curved, and the corner coordinates carried image-extraction error. This paper addresses that shortcoming. The boundary in closed form is derived and it is shown that the curved edge is an arc of an ellipse. Formulae for exact corner positions are given. A polygon form transition from five sided to three sided in the boundary at $H^* = \log_2 \varphi$ is identified. A set of integer comparator constants are derived whose rounding direction ensures the resulting test error is one-sided and so is conservative.

2 Count Coordinates and the Inverse Map

For this calculation the coordinates are normalized over a block of n bits with $N = n - 1$ bit-pair positions:

$$x = pc1 = \frac{count1}{N} = \hat{\mu}, \qquad y = pc11 = \frac{count11}{N}.$$

For the two-state Markov chain, y estimates $\Pr[x_i{=}1, x_{i+1}{=}1] = \mu P_{11}$ and x estimates μ, so the four transition probabilities can be recovered from (x, y) as

$$P_{11} = \frac{y}{x}, \qquad P_{10} = 1 - \frac{y}{x}, \qquad P_{01} = \frac{x - y}{1 - x}, \qquad P_{00} = \frac{1 - 2x + y}{1 - x}. \tag{1}$$

These are the empirical conditional transition frequencies: $x - y$ is *pc10* and $1 - 2x + y$ is *pc00* (each to within one count, since *count01* and *count10* can differ by at most one in non-cyclic data), so $P_{11} = count11/count1$ and $P_{00} = count00/count0$.

3 The Min-Entropy Rate in Closed Form

The most common value (MCV) of a b-bit symbol from the model is a repeating two-bit pattern (all ones, all zeros, or alternating), possibly with a deviating final bit for even b [1]. The symbol probabilities of the MCV candidates have the form (prefactor) $\cdot\, P_{11}^{b-1}$, (prefactor) $\cdot\, P_{00}^{b-1}$, or (prefactor) $\cdot\, (P_{01}P_{10})^{\lfloor (b-1)/2 \rfloor}$, where the prefactors such as $(P_{01} + P_{11})$ do not grow with b. The per-bit min-entropy rate is therefore determined by the stepwise geometric growth of the bit pattern of the MCV:

$$H_\infty(x, y) = -\log_2 \max\Big(P_{11},\ P_{00},\ \sqrt{P_{01}P_{10}} \Big) = -\log_2 \max\left(\frac{y}{x},\ \frac{1 - 2x + y}{1 - x},\ \frac{x - y}{\sqrt{x(1 - x)}} \right). \tag{2}$$

Equation (2) is the exact analytic form of the contours plotted in Figure 10 of [1]. The prefactors shift the finite-width boundary by $O(1/b)$; for a block-level online test the rate is the right object, and the finite-width correction is discussed in Section 10.

4 The Exact Boundary

Let $p = 2^{-H}$ for a chosen threshold H. The region $\mathcal{R}_H = \{(x, y) : H_\infty(x, y) \geq H\}$ is the intersection of three entropy constraints and the parameter feasibility constraints $0 \leq P_{01} \leq 1$, $P_{11} \geq 0$:

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Constraint	In (x, y)	Shape	Edge name
$P_{00} \le p$	$y \le (2-p)\,x - (1-p)$	straight line	AB
$P_{11} \le p$	$y \le p\,x$	straight line through origin	BC
$P_{01} \le 1$	$y \ge 2x - 1$	straight line	CD
$\sqrt{P_{01}P_{10}} \le p$	$(x-y)^2 \le p^2\,x(1-x)$	**ellipse**	DE
$P_{11} \ge 0$	$y \ge 0$	axis	EA

Proposition 1 (The curved edge is an ellipse arc)**.** *The constraint $(x-y)^2 \le p^2 x(1-x)$ expands to $(1+p^2)x^2 - 2xy + y^2 - p^2 x \le 0$, a conic with discriminant $B^2 - 4AC = 4 - 4(1+p^2) = -4p^2 < 0$: an ellipse. The boundary edge is the lower branch*

$$y = x - p\sqrt{x(1-x)}. \tag{3}$$

Since $\sqrt{x - x^2}$ is concave, $x - p\sqrt{x - x^2}$ is convex. The DE edge is therefore curved and convex, not piecewise linear, confirming the visual impression in [1]. Chords of a convex arc lie above it, so linear approximations by chords are automatically conservative (Section 6).

Notably, the CD edge $y \ge 2x - 1$ used in [1] was written down from the plot but is in fact *exact*: it is the feasibility line $P_{01} = 1$, equivalently *count00* $= 0$ to within one count.

5 Corners and the Two Regimes

Intersecting adjacent constraints yields closed forms for every corner. Two regimes arise, separated by whether the constraints $P_{11} = p$ and $P_{01}P_{10} = p^2$ can be simultaneously active with $P_{01} \le 1$: that requires $P_{01} = p^2/(1-p) \le 1$, i.e. $p^2 + p - 1 \le 0$, i.e.

$$p \ge \frac{\sqrt{5}-1}{2} = \frac{1}{\varphi} \quad \Longleftrightarrow \quad H \le H^* = \log_2 \varphi \approx 0.6942\,. \tag{4}$$

The intersections are worked through below, followed by the regime condition derived directly in (x, y) and a cross-check in Markov parameters.

5.1 Derivation of the corners

Write the boundary curves of Section 4 as functions of x:

$$\ell_{AB}(x) = (2-p)\,x - (1-p), \qquad \ell_{BC}(x) = p\,x, \qquad \ell_{CD}(x) = 2x - 1, \qquad y_{\text{arc}}(x) = x - p\sqrt{x(1-x)},$$

together with the axis $y = 0$. Throughout, $0 < p < 1$ and $0 \le x \le 1$. Each corner is the point where two adjacent curves are equal.

Line–line corners. These are linear equations in x:

$$\begin{aligned}
A:&\quad \ell_{AB}(x) = 0 && \Rightarrow\ (2-p)\,x = 1-p && \Rightarrow\ A = \Big(\frac{1-p}{2-p},\ 0\Big),\\
B:&\quad \ell_{AB}(x) = \ell_{BC}(x) && \Rightarrow\ 2(1-p)\,x = 1-p && \Rightarrow\ B = \Big(\frac{1}{2},\ \frac{p}{2}\Big),\\
C:&\quad \ell_{BC}(x) = \ell_{CD}(x) && \Rightarrow\ (2-p)\,x = 1 && \Rightarrow\ C = \Big(\frac{1}{2-p},\ \frac{p}{2-p}\Big).
\end{aligned}$$

The cancellation of $1-p$ in B is why the apex lies on $x = \frac{1}{2}$ at every threshold.

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Line–arc corners. Setting a line equal to y_{arc} and moving everything except the square root to one side gives an equation of the form $L(x) = p\sqrt{x(1-x)}$. Squaring gives a quadratic in x. The full ellipse $(x-y)^2 = p^2x(1-x)$ passes through $(0,0)$ and $(1,1)$, and each of the four boundary lines passes through one of these two points (ℓ_{BC} and $y=0$ through $(0,0)$; ℓ_{AB} and ℓ_{CD} through $(1,1)$). So every quadratic has a known root at $x=0$ or $x=1$. Factoring that root out leaves a linear equation whose root is the corner. In each case below, $L(x) \geq 0$ on $[0,1]$, so squaring introduces no sign-spurious roots. Let $\Delta = (1-p)^2 + p^2$.

Corner E ($y_{\text{arc}} = 0$, discard $x = 0$):

$$x = p\sqrt{x(1-x)} \;\Rightarrow\; x^2 = p^2x(1-x) \;\Rightarrow\; x\big[(1+p^2)\,x - p^2\big] = 0 \;\Rightarrow\; E = \Big(\frac{p^2}{1+p^2},\, 0\Big).$$

Corner D ($y_{\text{arc}} = \ell_{CD}$, discard $x = 1$):

$$\begin{aligned} x - p\sqrt{x(1-x)} = 2x-1 \;&\Rightarrow\; 1-x = p\sqrt{x(1-x)} \;\Rightarrow\; (1-x)^2 = p^2x(1-x) \\ &\Rightarrow\; (1-x)\big[1-(1+p^2)\,x\big] = 0 \\ &\Rightarrow\; x_D = \frac{1}{1+p^2}, \quad y_D = 2x_D - 1 = \frac{1-p^2}{1+p^2}. \end{aligned}$$

Corner C' ($y_{\text{arc}} = \ell_{BC}$, discard $x = 0$):

$$\begin{aligned} x - p\sqrt{x(1-x)} = p\,x \;&\Rightarrow\; (1-p)\,x = p\sqrt{x(1-x)} \;\Rightarrow\; (1-p)^2x^2 = p^2x(1-x) \\ &\Rightarrow\; x\big[\Delta\,x - p^2\big] = 0 \\ &\Rightarrow\; x_{C'} = \frac{p^2}{\Delta}, \quad y_{C'} = p\,x_{C'} = \frac{p^3}{\Delta}. \end{aligned}$$

Corner A' ($y_{\text{arc}} = \ell_{AB}$, discard $x = 1$):

$$\begin{aligned} x - p\sqrt{x(1-x)} = (2-p)\,x - (1-p) \;&\Rightarrow\; p\sqrt{x(1-x)} = (1-p)(1-x) \\ &\Rightarrow\; p^2x(1-x) = (1-p)^2(1-x)^2 \\ &\Rightarrow\; (1-x)\big[\Delta\,x - (1-p)^2\big] = 0 \;\Rightarrow\; x_{A'} = \frac{(1-p)^2}{\Delta}, \\ y_{A'} = \ell_{AB}(x_{A'}) = \frac{(1-p)\big[(2-p)(1-p) - \Delta\big]}{\Delta} &= \frac{(1-p)(1-p-p^2)}{\Delta}. \end{aligned}$$

The discarded roots are not corners of the region. On the axis the arc lies below $y=0$ for $0 < x < x_E$, because $x < p\sqrt{x(1-x)} \iff x < x_E$. Near $x=1$ the arc lies below $y = 2x-1$ for $x > x_D$, because $p\sqrt{x(1-x)} > 1-x \iff x > x_D$. In both intervals the arc is infeasible, and the line bounds the region instead. The pairs (x_A, x_C), (x_E, x_D) and $(x_{A'}, x_{C'})$ each sum to 1, as the complement symmetry requires.

The regime condition. On the axis $y=0$ the arc constraint $(x-y)^2 \leq p^2x(1-x)$ reads $x \leq x_E$, and the AB constraint reads $x \geq x_A$. The axis therefore contributes an edge EA exactly when $x_A \leq x_E$:

$$\frac{1-p}{2-p} \leq \frac{p^2}{1+p^2} \iff (1-p)(1+p^2) \leq p^2(2-p) \iff 1-p+p^2-p^3 \leq 2p^2-p^3 \iff 1-p-p^2 \leq 0,$$

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which is $p \geq 1/\varphi$. By the complement symmetry, the same condition makes $x_D \leq x_C$, so it also governs whether edge CD exists. When $1-p-p^2 > 0$, the line AB meets the arc before reaching the axis. Its meeting point A' has height $y_{A'} = (1-p)(1-p-p^2)/\Delta$, which is positive for exactly the same values of p. At the transition $p = 1/\varphi$ we have $1-p = p^2$, so $\Delta = p^4 + p^2$ and $x_{A'} = p^4/(p^2(1+p^2)) = x_E$: the corners A, E and A' coincide, and likewise C, D and C'. The boundary deforms continuously from one regime to the other.

Apex B stays inside the region for every $H < 1$. At $x = \frac{1}{2}$ the arc constraint requires $y \geq \frac{1-p}{2}$, and $y_B = \frac{p}{2}$ satisfies this iff $p \geq \frac{1}{2}$.

Cross-check in Markov parameters. Each corner is a point where two constraints on the transition probabilities hold with equality. Using the stationary mean $x = \mu = P_{01}/(P_{01}+P_{10})$ and $y = \mu P_{11}$ gives the same coordinates independently:

Corner	Active constraints	Resulting (P_{01}, P_{10})
A	$P_{00} = p,\ P_{11} = 0$	$(1-p,\ 1)$
B	$P_{00} = p,\ P_{11} = p$	$(1-p,\ 1-p)$
C	$P_{11} = p,\ P_{01} = 1$	$(1,\ 1-p)$
D	$P_{01} = 1,\ P_{01}P_{10} = p^2$	$(1,\ p^2)$
E	$P_{11} = 0,\ P_{01}P_{10} = p^2$	$(p^2,\ 1)$
A'	$P_{00} = p,\ P_{01}P_{10} = p^2$	$(1-p,\ p^2/(1-p))$
C'	$P_{11} = p,\ P_{01}P_{10} = p^2$	$(p^2/(1-p),\ 1-p)$

For example, at D we get $x = 1/(1+p^2)$ and $y = x(1-p^2)$. At A' the condition $P_{10} = p^2/(1-p) \leq 1$ is again $p^2 + p - 1 \leq 0$, which recovers the regime boundary stated above.

5.2 Five-edge regime ($H \leq H^*$)

The boundary has five edges AB, BC, CD, DE (arc), EA with corners

$$A = \left(\tfrac{1-p}{2-p},\ 0\right), \qquad B = \left(\tfrac{1}{2},\ \tfrac{p}{2}\right), \qquad C = \left(\tfrac{1}{2-p},\ \tfrac{p}{2-p}\right),$$
$$D = \left(\tfrac{1}{1+p^2},\ \tfrac{1-p^2}{1+p^2}\right), \quad E = \left(\tfrac{p^2}{1+p^2},\ 0\right). \tag{5}$$

B always lies on $x = \frac{1}{2}$; A, C and E, D are placed symmetrically about $x = \frac{1}{2}$, a consequence of the $0 \leftrightarrow 1$ complement symmetry of the model, which maps $(x, y) \mapsto (1-x,\ 1-2x+y)$, swaps the P_{00} and P_{11} constraints, and fixes the alternating constraint. For $H = 0.4$ ($p = 2^{-0.4} \approx 0.757858$) the exact corners are $A = (0.1949, 0)$, $B = (0.5, 0.3789)$, $C = (0.8051, 0.6101)$, $D = (0.6352, 0.2704)$, $E = (0.3648, 0)$, against the image-extracted values $(0.2, 0)$, $(0.5, 0.4)$, $(0.8, 0.6)$, $(0.6, 0.2)$, $(0.4, 0)$ of [1]. Figure 1 shows the exact region.

5.3 Triangle regime ($H > H^*$)

For $p < 1/\varphi$ the region detaches from both the $y = 0$ axis and the $P_{01} = 1$ line: corners C and D merge, E merges into A, and the region is bounded by the two entropy lines and the arc alone. With $\Delta = (1-p)^2 + p^2$:

$$A' = \left(\tfrac{(1-p)^2}{\Delta},\ \tfrac{(1-p)(1-p-p^2)}{\Delta}\right), \qquad B = \left(\tfrac{1}{2},\ \tfrac{p}{2}\right), \qquad C' = \left(\tfrac{p^2}{\Delta},\ \tfrac{p^3}{\Delta}\right), \tag{6}$$

as derived in Section 5.1, with $x_{A'} + x_{C'} = 1$. The constraints $y \geq 2x - 1$ and $y \geq 0$ remain true throughout the region but are implied by the others and never bind. Figure 2 shows the $H = 0.8$

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case, and Figure 3 the nested family across all nine thresholds, converging on the full-entropy point $(\frac{1}{2}, \frac{1}{4})$ as $H \to 1$.

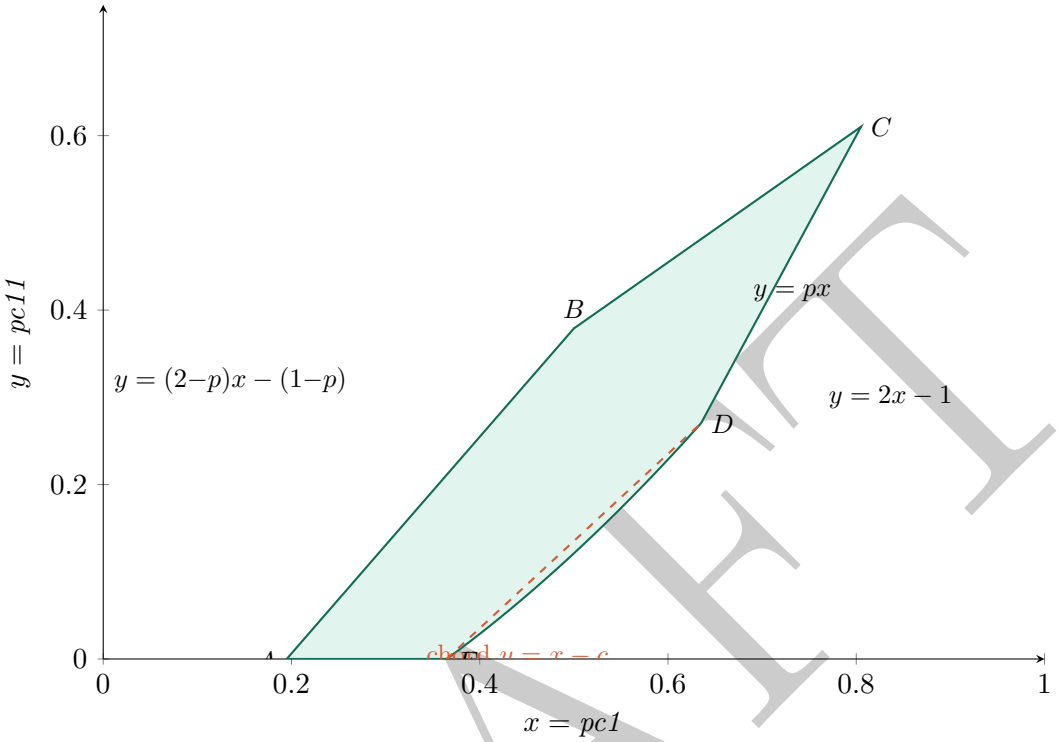


Figure 1: Exact region $\mathcal{R}_{0.4}$ ($p = 2^{-0.4}$): five-edge regime. The DE edge is the convex ellipse arc $y = x - p\sqrt{x(1-x)}$; the dashed line is the conservative slope-1 chord.

6 The Conservative Chord for the Elliptic Arc

Proposition 2 (Slope-1 chord)**.** *At every entropy level, the chord connecting the endpoints of the active arc has slope exactly* 1*. In the five-edge regime the endpoints are E and D and $D_x - E_x = \frac{1-p^2}{1+p^2} = D_y - E_y$; in the triangle regime the endpoints are A' and C' and the same cancellation occurs. The chord is*

$$y \;\geq\; x - c(H), \qquad c(H) = \begin{cases} \dfrac{p^2}{1+p^2}, & H \leq H^*, \\[2ex] \dfrac{p^2(1-p)}{(1-p)^2 + p^2}, & H > H^*. \end{cases} \tag{7}$$

Because the arc is convex and the chord endpoints lie on it, the chord lies above the arc: requiring $y \geq x - c$ is strictly stronger than the exact arc constraint, so replacing the arc with the chord yields a test that can only reject data whose true entropy is $\geq H$, never accept data below it. The chord is the only place the linear test gives anything away; the maximum sagitta (at $x = \frac{1}{2}$, five-edge regime) is $c(H) - \frac{1-p}{2}$, about 0.014 at $H = 0.4$.

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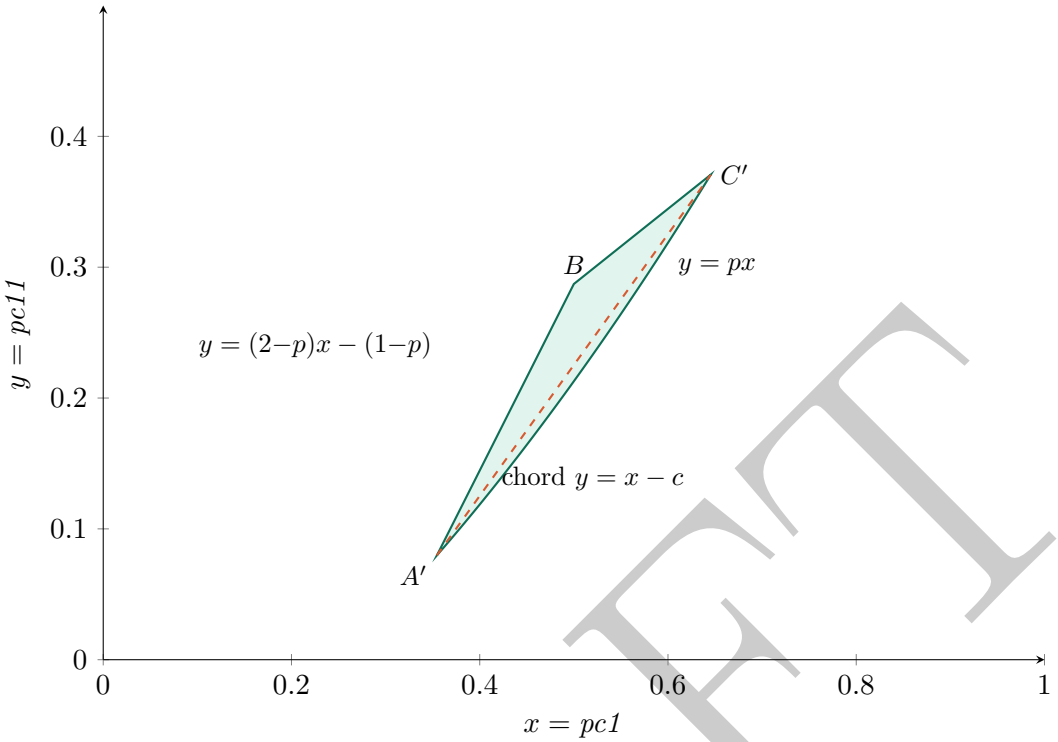


Figure 2: Exact region $\mathcal{R}_{0.8}$: triangle regime ($H > H^*$). The region has detached from $y = 0$ and from $y = 2x - 1$; only the two entropy lines and the arc bind.

7 Table of Linear Inequalities

Table 1 lists the four linear tests at nine entropy levels.

7.1 Perfect fit in the sub-threshold rows

For $H \leq H^*$ the chord line $y = x - c$ crosses $y = 0$ at $x = c = x_E$ exactly and crosses the DC line $y = 2x - 1$ at $x = 1 - c = x_D$ exactly. Consequently the five tests {AB, BC, DC, chord, $y \geq 0$} reproduce the exact corners A–E with no slack anywhere except along the arc itself. The entire conservatism of the linear polygon test is the thin sliver between the chord and the convex arc.

7.2 The $H = 0.5$ special case

At $H = \frac{1}{2}$, $p^2 = \frac{1}{2}$ exactly, so $c = 1/3$ and $D = (\frac{2}{3}, \frac{1}{3})$: the chord test becomes $3\,count11 \geq 3\,count1 - N$, pure shift-and-add with zero coefficient quantization. Moreover the *exact* arc test is dyadic at this level: $2(count1 - count11)^2 \leq count1\,(N - count1)$, with no approximation at all. $H = 0.5$ is the natural threshold for a 6:1-extraction design where hardware cost is at an absolute premium.

7.3 Conversion to counts and rounding direction

Multiplying each inequality by N converts $x \to count1$, $y \to count11$. All four constraints become monotonically *stricter* as p (or c) decreases: for AB, $\partial_p[(2-p)x - (1-p)] = 1 - x > 0$; BC and the chord are immediate. Therefore quantizing every irrational coefficient by rounding *down* (and

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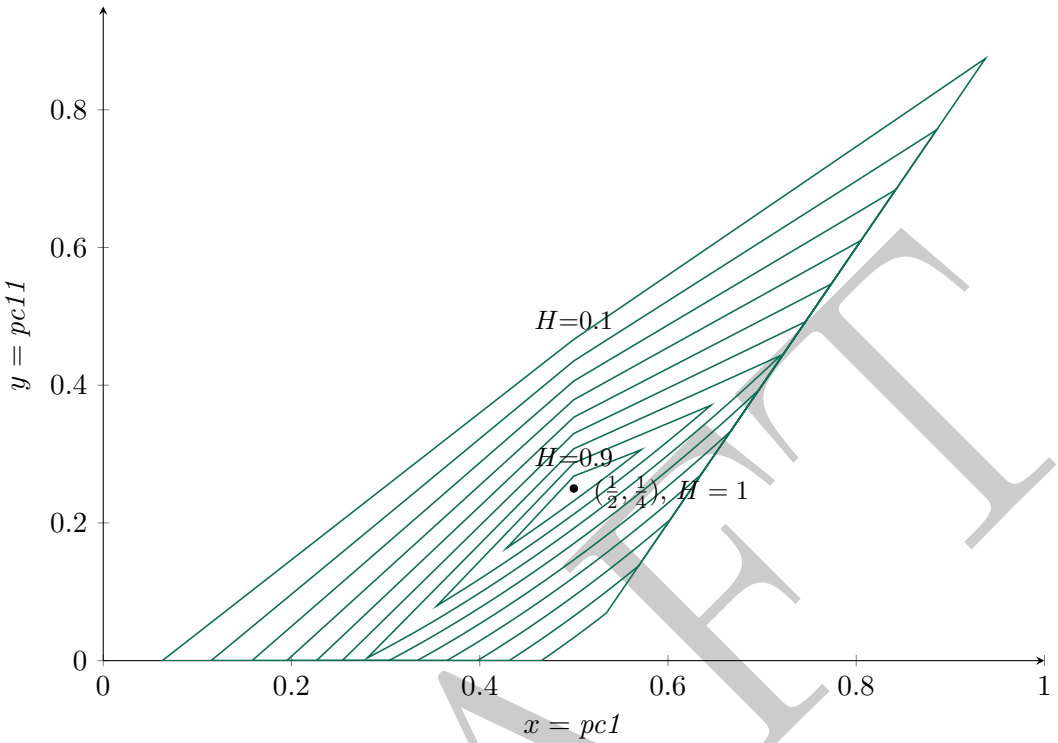


Figure 3: The nested family of exact boundaries for $H = 0.1, 0.2, \ldots, 0.9$, shrinking onto the full-entropy point $(\frac{1}{2}, \frac{1}{4})$. Apex B of each region lies on $x = \frac{1}{2}$ at height $p/2$; the structural transition from five edges to three occurs between $H = 0.6$ and $H = 0.7$, at $H^* = \log_2 \varphi$.

Table 1: Linear inequalities defining the conservative polygon at each entropy threshold, with $p = 2^{-H}$. The DC test is only an active boundary edge for $H \leq 0.6$ of the listed levels; above H^* it is implied by the others (marked —) but remains harmless to include.

H	$p = 2^{-H}$	AB: $y \leq$	BC: $y \leq$	DC: $y \geq$	ED chord: $y \geq$
0.1	0.933033	$1.066967x - 0.066967$	$0.933033x$	$2x - 1$	$x - 0.465398$
0.2	0.870551	$1.129449x - 0.129449$	$0.870551x$	$2x - 1$	$x - 0.431126$
0.3	0.812252	$1.187748x - 0.187748$	$0.812252x$	$2x - 1$	$x - 0.397501$
0.4	0.757858	$1.242142x - 0.242142$	$0.757858x$	$2x - 1$	$x - 0.364817$
0.5	0.707107	$1.292893x - 0.292893$	$0.707107x$	$2x - 1$	$x - 1/3$ (exact)
0.6	0.659754	$1.340246x - 0.340246$	$0.659754x$	$2x - 1$	$x - 0.303269$
0.7	0.615572	$1.384428x - 0.384428$	$0.615572x$	—	$x - 0.276565$
0.8	0.574349	$1.425651x - 0.425651$	$0.574349x$	—	$x - 0.274750$
0.9	0.535887	$1.464113x - 0.464113$	$0.535887x$	—	$x - 0.265198$

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the subtracted intercept term up) shrinks the accept region and guarantees one-sided behavior: the integer test can only reject data whose true entropy is at least H.

7.4 Tight triangle cases

At high thresholds the region is a thin triangle with apex $B = (\frac{1}{2}, p/2)$; at $H = 0.9$ the arc's sagitta is only $\approx$0.003, so the chord-versus-arc distinction becomes negligible, but coefficient quantization dominates: the fixed-point resolution of p must be fine enough that the rounded-inward triangle does not collapse to the empty set. Twelve fractional bits suffice for $N = 2303$ at $H = 0.9$ (Section 8); as $H \to 1$ the region converges to the single point $(\frac{1}{2}, \frac{1}{4})$, i.e. $count1 = N/2$, $count11 = N/4$, as it must for full entropy data.

8 Deriving the Integer Comparator Constants

Fix a block size n (hence $N = n - 1$) and a fixed-point scale $S = 2^k$. The four comparators, in two's complement arithmetic, are

$$\text{AB:}\quad 2^k\, count11 \;\le\; A_1\, count1 - A_0, \qquad A_1 = \lfloor (2-p)\, S \rfloor,\;\; A_0 = \lceil (1-p)\, S\, N \rceil, \tag{8}$$

$$\text{BC:}\quad 2^k\, count11 \;\le\; B_1\, count1, \qquad B_1 = \lfloor p\, S \rfloor, \tag{9}$$

$$\text{DC:}\quad count11 \;\ge\; 2\, count1 - N, \qquad \text{(exact; active for } H \le H^*\text{)}, \tag{10}$$

$$\text{ED:}\quad count11 \;\ge\; count1 - C_0, \qquad C_0 = \lfloor c(H)\, N \rfloor. \tag{11}$$

The multiplications by 2^k are realized in hardware as k-bit left shifts, i.e. wiring, at zero gate cost. The chord needs no fixed point at all because its slope is exactly 1. Table 2 gives the constants for the $n = 2304$ example of [1] ($N = 2303$, $k = 12$, $S = 4096$).

Table 2: Integer comparator constants for $n = 2304$, $k = 12$. Note $A_1 + B_1 = 8191$ in every row: the two slopes are $2 - p$ and p, both floored, so they straddle $2S$ by exactly one LSB. C_0 at $H = 0.5$ is $\lfloor 2303/3 \rfloor = 767$, the dyadic case.

H	A_1	A_0	B_1	C_0	DC active?	bitwidth
0.1	4370	631,706	3821	1071	yes	20
0.2	4626	1,221,108	3565	992	yes	21
0.3	4865	1,771,040	3326	915	yes	21
0.4	5087	2,284,145	3104	840	yes	22
0.5	5295	2,762,888	2896	767	yes	22
0.6	5489	3,209,571	2702	698	yes	22
0.7	5670	3,626,342	2521	636	implied	22
0.8	5839	4,015,202	2352	632	implied	22
0.9	5997	4,378,022	2194	610	implied	23

Bit widths: the left side 2^{12} *count11* reaches $2303 \cdot 4096 = 9{,}433{,}088$ (24 bits); the AB right side peaks near the same value and reaches $-A_0 \approx -4.4$M for small *count1*, so a 25-bit two's complement comparison covers all rows. The constant multiplies A_1 *count1* and B_1 *count1* synthesize to roughly 5–7 shift-adds each in canonical-signed-digit form; DC is a shift and the chord a subtract, keeping the full four-threshold test in the same sub-1000-gate class as the original polygon test [1], now with provably one-sided thresholds.

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As a spot check of tightness at the hardest level: at $count1 = 1152$ ($x \approx 0.5$), $H = 0.9$ yields the integer accept window $count11 \in [542, 617]$ against the exact real-valued window [540.7, 617.1] — one count of conservatism at the bottom, zero at the top.

Listing 1 generates the constants for arbitrary (H, n, k) with the correct rounding directions. One numerical caveat: double-precision evaluation of p leaves $\sim 2^{-52}$ of ambiguity, which matters only if a product lands within that distance of an exact integer; none of the nine rows above does, but for arbitrary parameters the floor/ceil operations should be evaluated with guard digits (e.g. `mpmath` or MPFR) to make them unambiguous.

Listing 1: Generator for the integer comparator constants.

```
import math

PHI_INV = (math.sqrt(5) - 1) / 2 # golden ratio threshold p* = 1/phi

def polygon_constants(H, n, k=12):
    N, S = n - 1, 1 << k
    p = 2.0 ** (-H)
    p2 = 2.0 ** (-2 * H)
    if p >= PHI_INV: # H <= log2(phi) ~ 0.6942
        c = p2 / (1.0 + p2) # chord intercept, arc endpoints E,D
        dc_active = True
    else: # triangle regime: arc endpoints A',C'
        c = p2 * (1.0 - p) / ((1.0 - p) ** 2 + p2)
        dc_active = False
    a0 = math.ceil((1.0 - p) * S * N)
    return {
        'A1': math.floor((2.0 - p) * S), # AB: (count11<<k) <= A1*count1 - A0
        'A0': a0,
        'B1': math.floor(p * S), # BC: (count11<<k) <= B1*count1
        'C0': math.floor(c * N), # ED: count11 >= count1 - C0
        'DC': dc_active, # DC: count11 >= 2*count1 - N
        'BW': math.ceil(math.log(a0,2))+1 # required bitwidth for arithmetic
    }

for H in [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]:
    print(H, polygon_constants(H, 2304))
```

9 Multiplier-Based Exact Test of the Elliptic Arc

Where two multipliers are affordable, the arc can be tested exactly rather than by chord, buying back the ∼1-count conservatism. The constraint $(x - y)^2 \le p^2 x(1 - x)$ scales to counts as

$$2^k \big(count1 - count11\big)^2 \;\le\; P_2 \cdot count1 \cdot \big(N - count1\big), \qquad P_2 = \lfloor p^2\, 2^k \rfloor, \tag{12}$$

where flooring P_2 is the conservative direction (smaller p^2 shrinks the region). For $k = 12$ the constants are $P_2 = 3565, 3104, 2702, 2352, 2048, 1783, 1552, 1351, 1176$ for $H = 0.1$–0.9; the $H = 0.5$ value is exactly 2048, reducing the constant multiply to a shift with zero quantization loss. Note $count1 - count11$ is simply $count10$, so the left operand is directly available if a 10-pair counter exists.

For $n = 2304$: $d = count1 - count11 \le 2303$ (12 bits), $d^2 \le 5.3$M (23 bits), left side $2^{12} d^2$ needs 35 bits; $count1(N - count1) \le \lfloor N^2/4 \rfloor \approx 1.33$M (21 bits), and $P_2 \cdot q \le 4096 \cdot 1.33$M (33 bits). A single 35-bit unsigned comparison completes the test. The datapath (Figure 4) is two subtractors,

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one 12×12 squarer, one 12×12 multiplier, one constant multiplier (CSD shift-add network, or a true multiplier if H must be runtime-programmable), and a comparator. The test fires once per n-bit block, so a single multiplier can be time-shared across the three products at three cycles per block if area is critical.

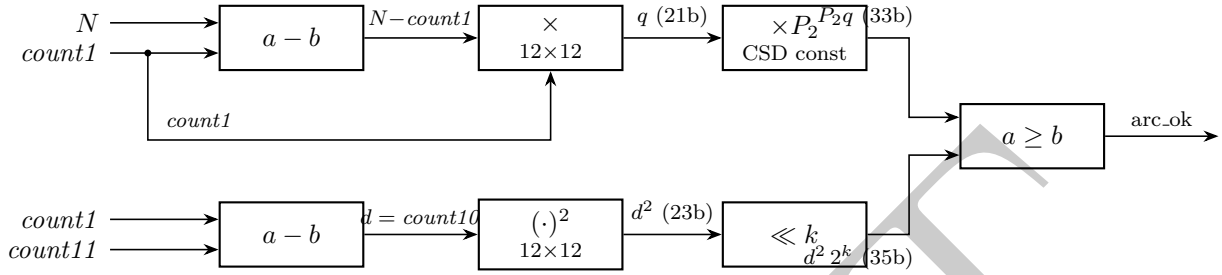


Figure 4: Datapath for the exact elliptic-arc test $2^k d^2 \le P_2 \cdot count1 \cdot (N - count1)$ with $d = count1 - count11 = count10$ and $P_2 = \lfloor p^2 2^k \rfloor$: two subtractors, one squarer, one multiplier, one constant (CSD) multiplier, and a 35-bit comparator; the $\ll k$ block denotes the k-bit left shift realizing the 2^k scaling as wiring. The test fires once per block, so the two array multipliers may be time-shared. At $H = 0.5$, $P_2 = 2^{k-1}$ and the constant multiplier reduces to a wire shift.

10 Drawbacks and Scope

The limitations of [1] are inherited unchanged: the test applies to single-bit-symbol sources whose statistics are captured by mean and serial correlation, i.e. sources well modeled by the two-parameter Markov chain; sources with longer-range structure (e.g. ring oscillators with complex phase noise coupling) require different methods. The boundary derived here is the *asymptotic* rate boundary; for certification of finite symbol widths b, the same three-pattern maximum applies with the full MCV prefactors of [1], shifting each edge by $O(1/b)$. Finally, the counts over a finite block are estimates of (x, y); the statistical false-positive rate of the online test at a given block size is set by the sampling distribution of the counts near the boundary, exactly as for the graphically derived polygon, and is unchanged by this work — what changes is that the deterministic component of the error (boundary placement) is eliminated.

11 Conclusion

A complete closed for description of the polygon test boundary from [1] is derived.

The region of $(count1, count11)$ pairs with min-entropy rate at least H is bounded by the two lines $y \le px$ and $y \le (2-p)x - (1-p)$, the line $y \ge 2x - 1$ when $H \le \log_2 \varphi$) and one curved edge that is a convex arc of an ellipse $(x - y)^2 = p^2 x(1 - x)$.

All corners have closed forms. A chord used as an easily computed approximation to the arc has slope exactly 1 with intercept $c(H)$ given in equation (7).

An efficient two multiplier structure is given to identify points inside the arc but outside the chord approximation.

By rounding integer constants toward the interior of the polygon yields a hardware test with only one-sided error. The uncertainty inherent in the image-extraction used in [1] is eliminated while not increasing the gate count of digital hardware implementations of the test.

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References

[1] Johnston, D.: Online Testing Entropy and Entropy Tests with a Two State Markov Model. In: Proceedings of SPACE 2024. Springer (2024)

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